



Modeling flow pattern transitions in electrical submersible pump under gassy flow conditions

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ABSTRACT

Gas entrainment is frequently encountered in electrical submersible pumps (ESP). When occurring, ESP suffers from moderate to severe performance degradation accompanied by unstable operations, depending on the inlet gas volumetric fraction (GVF) and pump rotational speed. The resulted pressure surging and instabilities may cause pump vibrations and short run-life. For a better design of ESP-based production system, the mechanistic model is needed to predict its performance under gas-liquid flow conditions. Similar to multiphase pipe flow, the flow pattern identification and classification inside a rotating ESP is important. In this paper, a new mechanistic model is proposed, which can map different flow patterns inside ESP with gas presence.

Experimental results reveal that the boosting pressure of ESP degrades as GVF increases. With ESP's *H-Q* performance curves obtained, the flow patterns including dispersed bubble flow, bubbly flow, intermittent flow and segregated flow can be identified by deflection points on *H-Q* curves (Gamboa and Prado, 2008). The transition boundaries between different flow patterns are mapped. Starting from the free body diagram on a stable bubble in the rotating flow field, the transition boundaries of dispersed bubble flow to bubbly flow and bubbly flow to intermittent flow are formulated. Based on the combined momentum equation, the transition criterion of intermittent flow to segregated flow is derived. The flow pattern map predicted by mechanistic model agrees well with that detected from the experimental *H-Q* curves of ESP under gassy flow conditions.

1. Introduction

ESPs are widely used in offshore fields where high production rates are required (Zhu et al., 2017a and 2017b). The compactly assembled multistage ESP with good hydraulic efficiency satisfies offshore high production needs. However, ESP's performance suffers from severe degradation with gas presence (Zhu et al., 2019a), resulting in pump unstable operations, reductions of boosting pressure and production rates. Due to the complicated pump geometries and rotating effect, it is extremely difficult to observe and characterize the internal multiphase flow structures by direct visualizations (Zhu et al., 2019b).

Experimental studies on ESP boosting pressure with various flow conditions have been conducted extensively, including viscous fluid flow (Zhu et al., 2016; Zhang, 2017), oil/water emulsion flow (Banjar et al., 2013; Croce Mundo, 2014), gas/liquid two phase flow (Cirilo, 1998; Romero, 1999; Pessoa, 2001; Beltur, 2003; Duran, 2003; Zapata, 2003; Barrios, 2007; Gamboa, 2008; Zhu and Zhang, 2017), as well as

slurry flow (Russell and Marsis, 2013). The entrainment of free gas results in the degradation of pump's hydraulic head, which further causes pressure surging and gas pocket formation (Shi et al., 2018, 2019; Mansour et al., 2018). Worse still, it triggers the instabilities that are detrimental to pump's functionalities, and jeopardizes the ESP-based production system (Zhu and Zhang, 2018). Surging can induce pump vibrations, leading to ESP short service-life (Zhou and Sachdeva, 2010), while the formation of gas pockets reduces production rates of hydrocarbon fluids significantly (Takacs, 2009). High vibrational stress and unexpected operation conditions may even cause serious damage to pumping devices (Neumann et al., 2016).

The pioneering experimental studies on two-phase flow in a rotating centrifugal pump were conducted by Murakami and Minemura (1974a and 1974b). Since then, extensive experimental work focuses on measuring the bulk properties of a centrifugal pump with the multiphase flow through it, including boosting pressure, hydraulic efficiency, shaft torque etc. Few studies (Barrios, 2007; Gamboa, 2008; Barrios and

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Prado, 2011; Verde et al., 2017) are related to the visualization of two-phase flow patterns inside a rotating centrifugal pump, especially the impeller flow channels. However, the recognition and characterization of flow patterns are essential to the mechanistic model of multiphase flow (Shoham, 2006; Daraboina et al., 2018). Depending on the fluid properties, flow rates and ESP geometries, the flow patterns can vary since the phase interaction and interfacial momentum transfer differ significantly. Flow patterns can affect the multiphase flow characteristics, but also map out the flow regimes corresponding to severe performance degradation and unstable operations so that the ESP run-life can be extended by avoiding such flow conditions.

Due to the aforementioned difficulties, the direct observations of flow patterns inside a rotating ESP are extremely hard. It involves necessary modifications on pump geometries, such as the removal of impeller hub and attaching the Pyrex glass on its top to enable visualizations (Barrios, 2007; Verde et al., 2017). Alternatively, the special instrumentation of non-intrusive measurement is incorporated. For instance, the high-resolution computed tomography (HireCT) has been used to indirectly observe and measure the distribution of in-situ gas void fractions inside a rotating centrifugal pump (Schäfer et al., 2015; Neumann et al., 2016).

Besides the direct observations, the identification of flow pattern transitions in a rotating ESP can be achieved by detecting the deflection points on pump's performance curves. Such indirect approach was first proposed by Gamboa and Prado (2011a and 2011b), who also proposed empirical correlations to predict the transition boundaries between varying flow patterns. Fig. 1 sketches the procedure for recognizing and identifying different flow patterns by interpreting the H - Q performance curves in ESP under gassy flow conditions.

As shown in Fig. 1, Gamboa and Prado (2011a and 2011b) associated the deflection points on H - Q performance curves with the flow pattern transitions. Although they claimed that such analogy could merely achieve a rough accordance, it did offer significant convenience to study the two-phase flow characteristics in a rotating ESP. Fig. 1(a) and (b) present the schematic H - Q performance curves of an ESP tested at the constant Q_L and Q_G , corresponding to the so-called surging test and mapping test, respectively. Three schematic curves (a, b and c) in Fig. 1(a) are obtained by the continuous measurement of the boosting pressure of an ESP at three different Q_L with Q_G increasing from zero to the maximum, where the ESP performance approaches nil. Quantitatively, Q_L decreases from case (a) to (c), e.g. $Q_{L,a} > Q_{L,b} > Q_{L,c}$, which can also be verified by the vertical black-dashed-lines in Fig. 1(b). For Fig. 1(b), the blue line is the single-phase performance curve, while the red curve is a typical H - Q performance curve of ESP tested at constant

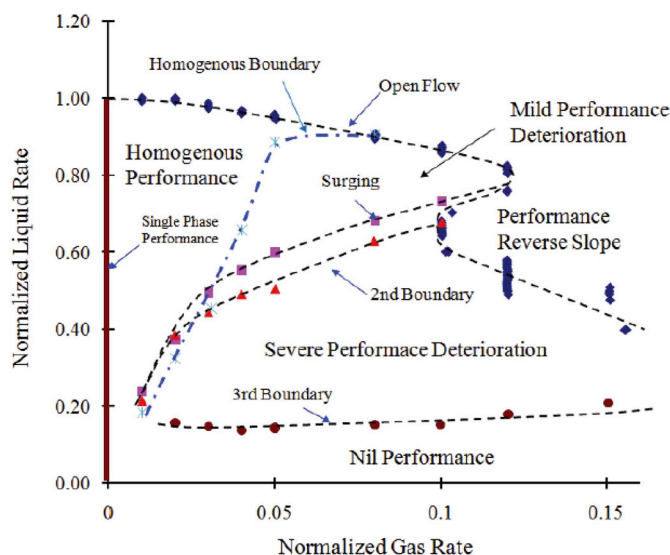


Fig. 2. Flow pattern map for ESP with gas/liquid two-phase flow by the performance-curve-detect approach (Gamboa and Prado, 2011a).

Q_G and varying Q_L . It is noted that different markers show the transitions of different flow patterns. The transition from dispersed bubble flow to bubbly flow is denoted by circles (○); the transition between bubbly flow and intermittent flow is shown by diamonds (◇); the intermittent flow and segregated flow are distinguished by squares (□). The triangles (△) marked on each section of the H - Q performance in Fig. 1(b) stand for different flow patterns, which are consistent with Fig. 1(a) plotting at a fixed Q_G as denoted by the vertical black-dashed-line.

Fig. 2 illustrates a typical flow pattern map in a rotating ESP under gassy flows (Gamboa and Prado, 2011a), which is summarized by diagnosing all the deflections on H - Q performance curves. As seen, four different flow regimes are mapped, including homogeneous performance, mild performance deterioration, severe performance deterioration, nil performance. The dashed-lines correspond to the transition boundaries between different flow regimes, which are directly obtained from visualization experiments. For smaller Q_G and larger Q_L , the homogeneous flow regime prevails inside a rotating ESP. As Q_G increases, the mild performance deterioration regime corresponding to bubbly flow dominates. The severe performance deterioration occurs at

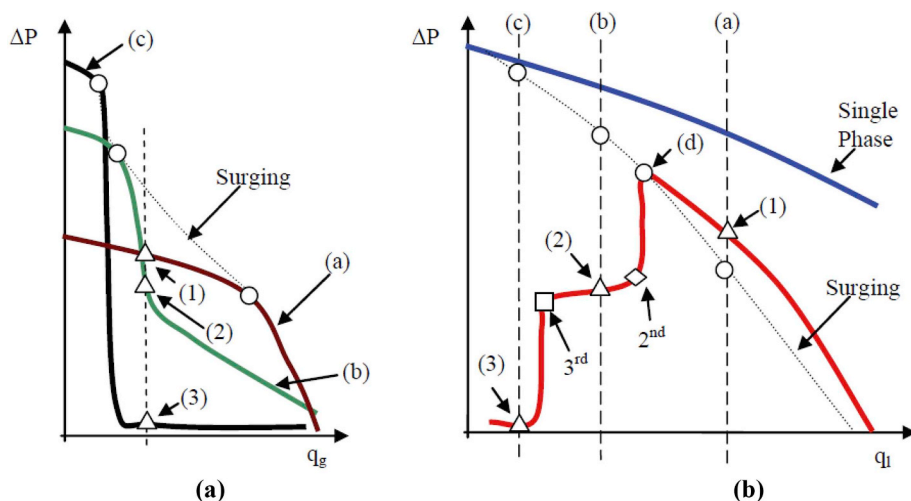


Fig. 1. Flow pattern recognition by interpreting performance curves (Gamboa and Prado, 2011a), (a) surging test performance curve, (b) mapping test performance curve.

higher Q_G and lower Q_L , featured by gas-pocket formation. Further increase of Q_G and decrease of Q_L may result in ESP nil performance, indicating the segregated flow pattern accompanied by gas locking.

Although many studies on ESP performance under gassy flow are available in literature, the gas bubble behavior and the effects on the two-phase flow hydrodynamics in rotating ESPs are still not well understood. Especially, the flow pattern recognition and identification, which are closely related to the mechanistic model of gas/liquid flow (Wang et al., 2019), are not clearly studied yet. Due to significant difficulties, the direct observations and visualizations of flow patterns inside rotating ESPs are very few. In this paper, a mechanistic model to map flow patterns in an ESP operated under gassy flow conditions is proposed. With the pump experimental H - Q performance curves, the flow patterns and transition boundaries can be mapped. Starting from the free body diagram on a stable single bubble, the transitions between dispersed bubble and bubbly flow, bubbly and intermittent flows are formulated. Based on the combined momentum equation, the transition criterion of intermittent to segregated flow is derived. The flow pattern map calculated from the new mechanistic model agrees well with that detected from ESP H - Q performance curves.

2. Experimental study

The experimental study has been elaborated in our previous work (Zhu et al., 2018). Built from Salehi (2012) flow loop for testing ESP two-phase performance, the new experimental facility consists of a fully closed liquid flow loop and an open loop for compressed air circulation, whose overall capacity is about 5000 bpd of water flow rate and 186 cfm of air flow rate at 217 psig. In this section, the experimental data is briefly presented, from which similar flow pattern maps to Gamboa and Prado (2011a and 2011b) study can be extracted and summarized. Then, the proposed mechanistic model can be validated by comparing against the obtained flow pattern maps.

The experiments were conducted with two testing schemes, namely surging and mapping tests. For surging tests, the liquid flow rate (Q_L) is fixed, while the gas flow rate (Q_G) increases continuously from zero to the maximum flow rate, at which the ESP approaches nil performance. For mapping tests, Q_G is fixed, whereas Q_L reduces from the maximum open rate to the minimum value corresponding to the shut-in of control valve. During both schemes, the stage pressure of ESP was recorded accordingly.

3. Surging test results

Fig. 3 shows the surging test results at stage 3 with different separator pressure P_{sep} , which served to balance loop pressure and control the intake gas properties. The red dashed-lines connect similar deflection points from each plot and group the data points according to different flow regimes. The transition boundaries can then be obtained by diagnosing the deflections visually.

As can be seen, the ESP boosting pressure dramatically changes at a certain GVF, before which the pump performance suffers from mild deterioration. This phenomenon is termed as pressure surging in ESP, which was first proposed by Lea and Bearden (1982). It coincides with ESP's unstable operations and flow pattern transitions, which significantly affects ESP pressure-boosting abilities under gassy flow conditions (Zhu and Zhang, 2016).

A rough comparison among the subfigures indicates that the ESP boosting pressure decreases with the increase of inlet GVF consistently for varying P_{sep} . At higher P_{sep} , the ESP performance improves slightly, since the drastic drop of pump stage pressure increment initiates at larger GVF. This is explainable by the bubble/droplet break-up and coalesce mechanism in two immiscible phases (Hinze, 1955). When the intake pressure increases, the gas density becomes larger, which leads to a finer distribution of gas bubbles. Thus, the surface area to volume ratio of gas increases, resulting in a higher total surface free energy of

dispersed gas bubbles. The major disruptive force for gas bubbles is the turbulent force, which should be even larger to collide and break up the small bubbles so as to coalesce them and generate the bigger one. Without larger bubble formation (referred as Taylor bubble in multiphase pipe flow), the ESP performance will be more stable and less likely to suffer from drastic drop due to the gas-pocket formation or gas locking. Hence, the ESP performance in terms of boosting pressure is slight better as the system pressure goes higher. Similar trend is also observed in mapping test results, shown in Fig. 4.

4. Mapping test results

Fig. 4 shows the mapping test results at stage 3 with different P_{sep} . In each figure, the plot is based on Q_{gd} , a dimensionless gas flow rate defined as Q_G/Q_{max} , where Q_{max} is the open flow rate in ESP. Similar to Fig. 3, the red-dashed-lines correspond to deflection points from each plot, which separate the data points into different flow regimes. As seen, the ESP performance is affected by gas and liquid flow rates significantly. For relatively small Q_{gd} , the H - Q performance is similar to water curve, meaning that the boosting pressure increases monotonically as Q_L decreases.

However, the boosting pressure of ESP at slightly higher Q_{gd} is consistent with the water performance curve until a sudden breakdown. As Q_L reduces, the stage pressure increment in ESP suffers from degradation increasingly. At larger Q_{gd} , the breakdown occurs at higher Q_L , indicating that ESP's ability to handle gas-entrainment weakens. With the decrease of P_{sep} , the ESP pressure increment deviates from water performance curve at an even higher Q_L . Moreover, ESP boosting pressure after performance breakdown approaches nil quickly as Q_L decreases. As concluded by Gamboa (2008), the breakdown of ESP boosting pressure under gassy flow from water performance curve is due to the transition from dispersed bubble flow to bubbly flow. With Q_L decreases further, the intermittent flow pattern may occur, corresponding to the second turning point on the H - Q performance curves.

5. Mechanistic modeling

This section discusses the development of mechanistic model for predicting flow patterns in the rotating ESPs. By analogy to multiphase pipe flow, similar modeling methodology is applied here.

6. Slug dynamics in ESP

As shown in Fig. 5, the slug flow in pipe consists of a liquid slug body and a gas core region. For the slug flow in rotating ESP (see Fig. 5(b)), similar structures can be observed from previous visualization experiments, although the exact shapes of the Taylor bubbles may vary. In this study, the slug flow regime in a rotating ESP under gassy conditions is assumed to associate with Taylor bubble flow, which is consistent with slug flow in pipes.

Derived by Sachdeva (1998) and Sun (2003), the momentum equations along the streamline in a rotating centrifugal pump should be written as below

$$\left. \frac{dp}{dr} \right|_{streamline} = \left[-\rho_p W_p \frac{dW_p}{dr} + \rho_p \Omega^2 r + \left(\frac{dp}{ds} \right)_{f,p} \frac{ds}{dr} - \frac{M_{p,s}}{\alpha_p} \frac{ds}{dr} \right]_{streamline} \quad (1)$$

where the subscript p is for phase, e.g. g for gas and l for liquid phase; $streamline$ mean that the forces are projected on the streamline direction. $M_{p,s}$ is the interfacial momentum transfer between gas and liquid phases.

Multiplying Eq. (1) by dr/ds

$$\frac{dp}{ds} = \frac{-\rho_p W_p \frac{dW_p}{ds}}{b} + \frac{\left(\frac{dp}{ds} \right)_{f,p}}{c} - \frac{M_{p,s}}{\alpha_p d} + \frac{\rho_p \Omega^2 r}{e} \Big|_{streamline} \quad (2)$$

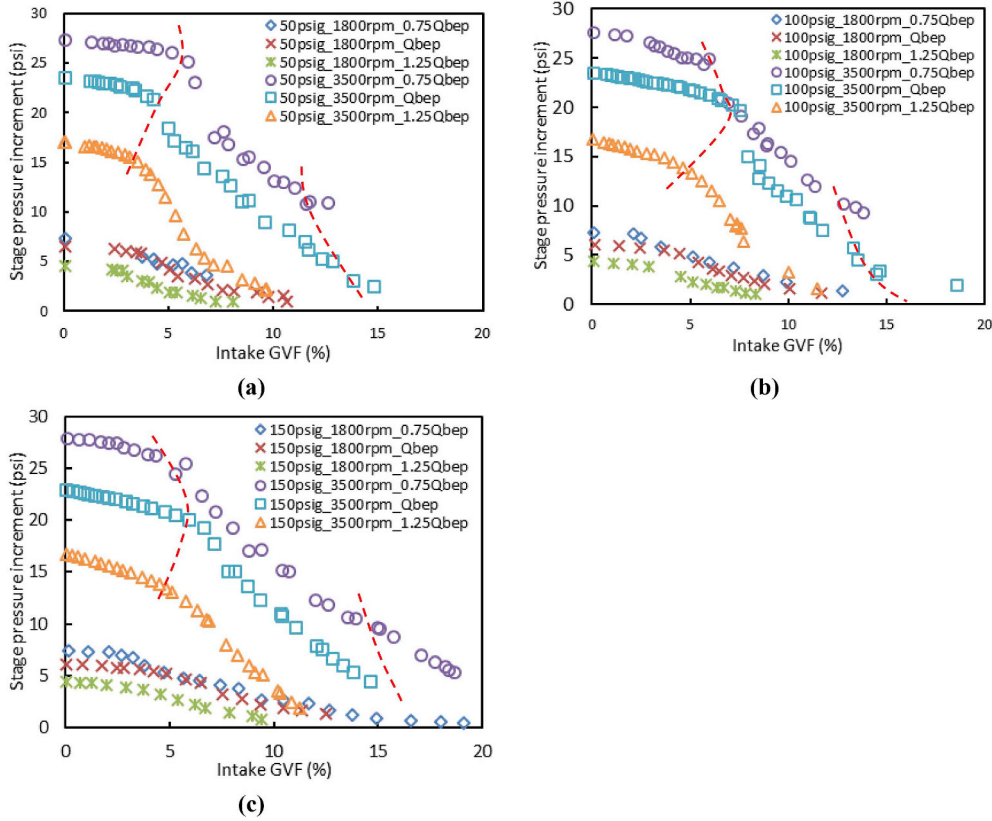


Fig. 3. Surging test results at stage 3 with different separator pressure (Zhu et al., 2018), (a) Psep = 50 psig, (b) Psep = 100 psig, (c) Psep = 150 psig.

where the subscript s is for the streamline direction. Terms $a \sim e$ in the above equation correspond to pressure gradient term, advection term, friction term, interfacial momentum transfer term, and centrifugal body force, term respectively. Apply Eq. (2) to the slug film in rotating centrifugal pump,

$$\frac{(p_2 - p_1)}{l_F} = \frac{\rho_L(v_T - v_F)(v_S - v_F)}{l_F} - \frac{\tau_I S_I + \tau_F S_F}{H_{LF} A} + \rho_L \Omega^2 R_I|_{streamline} \quad (3)$$

and

$$\frac{(p_2 - p_1)}{l_F} = \frac{\rho_C(v_T - v_C)(v_S - v_C)}{l_F} + \frac{\tau_I S_I}{(1 - H_{LF}) A} + \rho_C \Omega^2 R_I|_{streamline} \quad (4)$$

where Eq. (3) is the momentum balance of liquid phase, and Eq. (4) is for gas core/Taylor bubble. A is the cross-sectional area of ESP flow channel, v_T , v_F , v_S and v_C are the translational velocity, film velocity, slug body velocity, and gas core velocity, l_F is the liquid film length. H_{LF} is the film liquid holdup. S_I , S_F is the wet perimeters for interface and liquid film. ρ_L and ρ_C are liquid and gas core densities. Ω is the pump rotational speed and R_I is the representative radius of impeller. Subscripts 1 and 2 denote the front and tail of Taylor bubble. Then the combined momentum equation in a rotating ESP slug flow is obtained as

$$\frac{\rho_L(v_T - v_F)(v_S - v_F) - \rho_C(v_T - v_C)(v_S - v_C)}{l_F} - \frac{\tau_F S_F}{H_{LF} A} - \tau_I S_I \left(\frac{1}{H_{LF} A} + \frac{1}{(1 - H_{LF}) A} \right) + (\rho_L - \rho_C) \Omega^2 R_I|_{streamline} = 0. \quad (5)$$

Compared to the combined momentum equation for slug flow in pipe (Zhang et al., 2003a and 2003b), the difference in Eq. (5) is the body force term. All the velocities are the relative velocities to the ESP flow channel.

7. Flow pattern transition

The transition boundaries between flow patterns are similar to two-phase pipe flow. The flow map can be divided into four different flow regimes, namely dispersed bubble flow, bubbly flow, intermittent flow, and segregated flow. Based on the limited visualization results, the intermittent and segregated flows in a rotating ESP are slug flow and concurrent annular flow, respectively.

7.1. Dispersed bubble to bubbly flow transition

Compared to multiphase pipe flow, the transition boundary for dispersed bubble to bubbly flow is modeled as (Shoham, 2006):

$$d_{max} < d_{CD} \quad (6)$$

where d_{max} is maximum bubble size in turbulent flow field, d_{CD} is the critical bubble diameter at which the bubbles start to deform and coalesce to form bigger ones. This critical diameter was first proposed by Brodkey, (1967) and then modified by Barnea et al. (1982):

$$d_{CD} = 2 \left[\frac{0.4\sigma}{(\rho_L - \rho_G)g} \right]^{0.5} \quad (7)$$

In turbulent flow, bubbles are subjected to deformation, break-up, and coalescence by turbulence forces and interfacial tensions. According to Hinze (1955) theory, turbulence forces (τ), acting on the surface of the fluid particle tend to break up the bubbles into finer ones. On the contrary, the interfacial tensions tend to coalesce small bubbles to generate bigger ones.

Similar bubble breakup and coalescence mechanism in pipe flow can be applied to rotating ESP multiphase flow with necessary modification on the calculations of d_{max} and d_{CD} . The bubble size in a rotating ESP impeller can be estimated as (Zhu and Zhang, 2017):

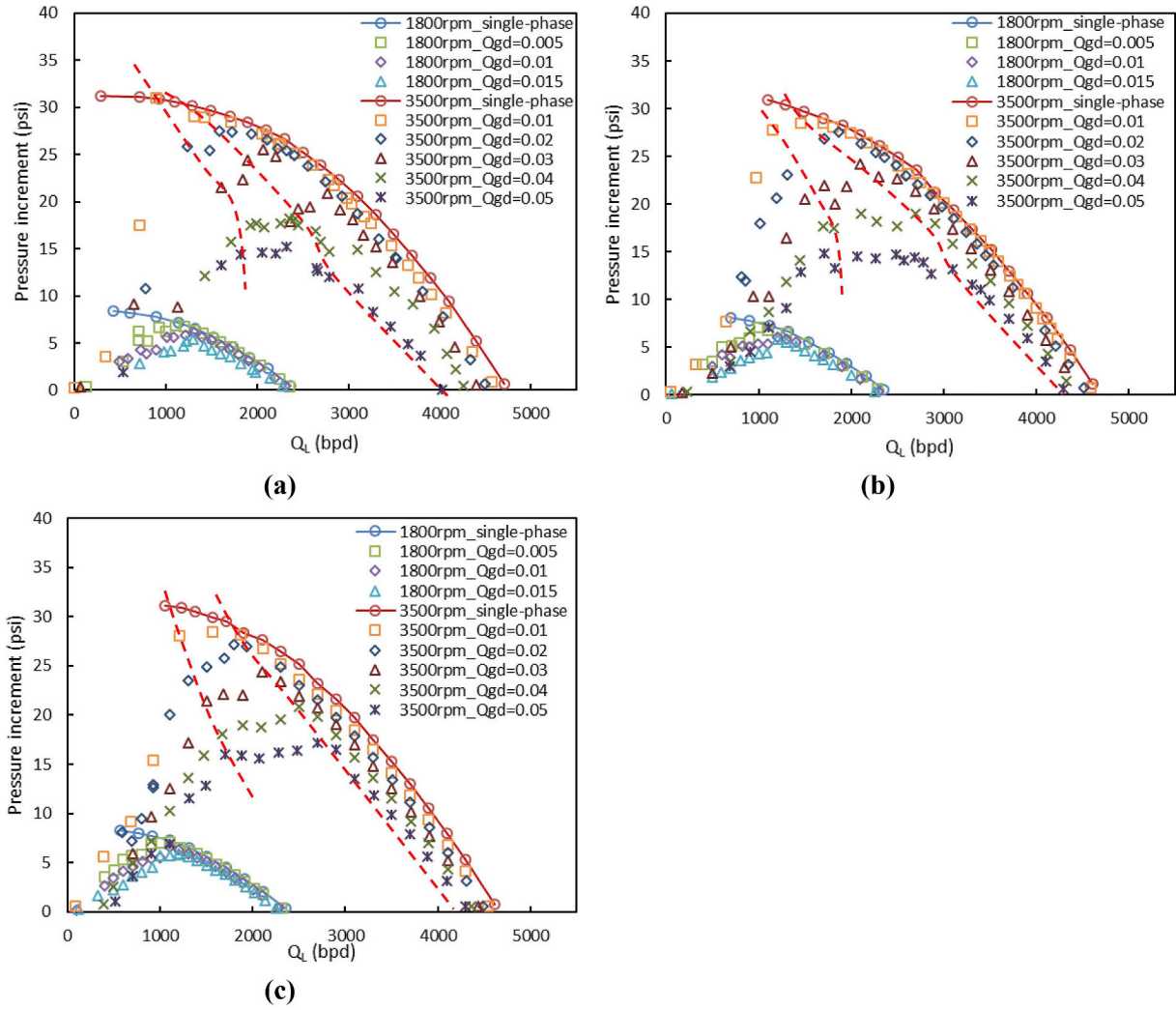


Fig. 4. Mapping test results at stage 3 with different separator pressure (Zhu et al., 2018), (a) Psep = 50 psig, (b) Psep = 100 psig, (c) Psep = 150 psig.

$$d_{max} = 10.056\lambda_G \left(\frac{\sigma}{\rho_c} \right)^{\frac{3}{5}} \left(\frac{\Delta P Q_L}{\rho_c V} \right)^{-\frac{2}{5}} \left(\frac{\rho_c}{\rho_d} \right)^{\frac{1}{5}} \quad (8)$$

where λ_G is GVF, σ is the surface tension, ΔP is the pressure increment of single stage ESP, Q_L is the liquid flow rate, and V is the impeller volume. Subscripts c and d denote the continuous and dispersed phases. Based on Barnea et al. (1982) study, the critical bubble diameter the centrifugal multiphase flow can be modified as:

$$d_{CRIT} = 2 \left[\frac{0.4\sigma}{(\rho_L - \rho_G)\Omega^2 R_I} \right]^{0.5} \quad (9)$$

where R_I is the representative impeller radius.

Substitute Eq. (8) and Eq. (9) into Eq. (6), the mechanistic model to predict the critical GVF at which ESP surging initiates can be obtained as:

$$\lambda_C = \frac{2 \left[\frac{0.4\sigma}{(\rho_L - \rho_G)\Omega^2 R_I} \right]^{\frac{1}{2}}}{10.056 \left(\frac{\sigma}{\rho_L} \right)^{\frac{3}{5}} \left(\frac{\Delta P Q_L}{\rho_L V} \right)^{-\frac{2}{5}} \left(\frac{\rho_L}{\rho_G} \right)^{\frac{1}{5}}} \quad (10)$$

7.2. Bubbly flow to intermittent flow

The transition for bubbly flow to slug flow in pipeline is predicted by bubble packing limit in three-dimensional cubic lattice. The bubble coalescence increases sharply promoting the formation of Taylor-bubbles and slugging if the void fraction reaches 0.25, e.g. $\alpha_{Crit} = 0.25$. In centrifugal multiphase flow, a new correlation for estimating α_{Crit} is proposed, which accounts for the rotating effect in ESPs:

$$\alpha_{Crit} = \frac{\pi}{6} - \left(\frac{\pi}{6} - \frac{1}{4} \right) \exp \left(- \left(\frac{N}{N_{REF}} \right)^n \right) \quad (11)$$

where N_{REF} is the rotational speed at best efficient point. $n = 4.0$ is used in this study.

Zhu and Zhang (2016) proposed a mechanistic model to calculate the in-situ gas void fraction in a rotating ESP impeller under gas/liquid two-phase flow. The model incorporates ESP geometries, fluid properties, and operation parameters, which can be applied to various flow conditions. The calculation formula is given by:

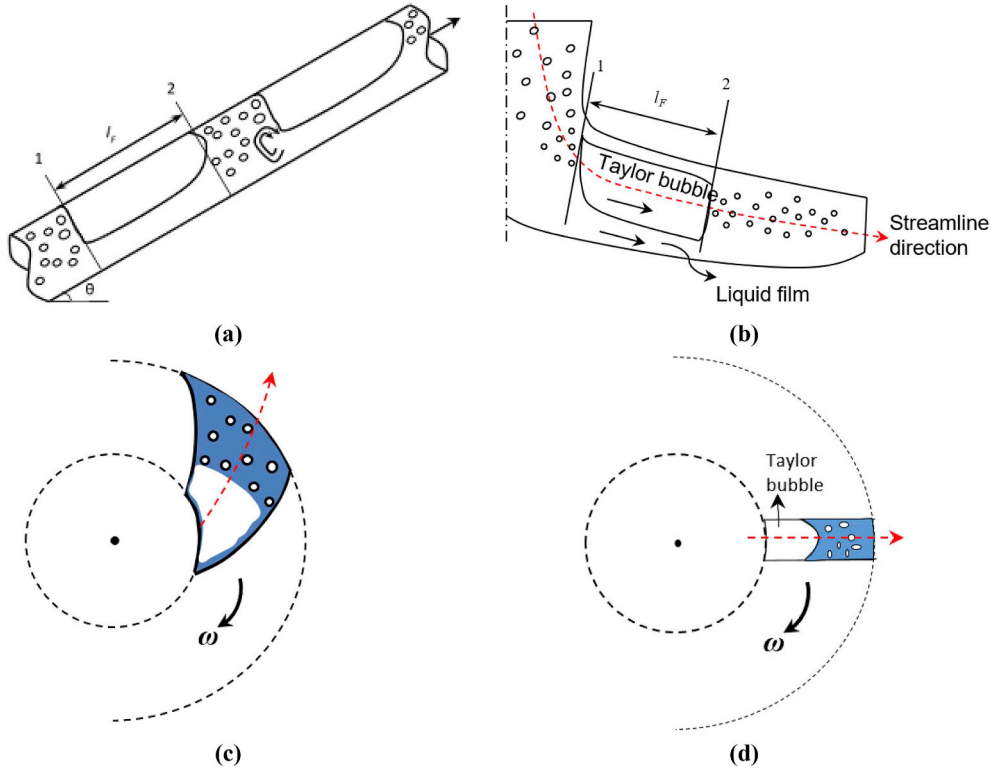


Fig. 5. Schematic of slug flows in pipe and centrifugal pump, (a) control volume in unified model for two-phase pipe flow, (b) and (c) slug flow structure in centrifugal pump, (d) assumed slug flow structure based on equivalent hydraulic parameters.

$$\alpha_G = \frac{R_S - 1 + \sqrt{(1 - R_S)^2 + 4R_S\lambda_G}}{2R_S} \quad (12)$$

where R_S is expressed by

$$R_S = \sqrt{\frac{4d_B(\rho_L - \rho_G)R_I\Omega^2(2\pi R_I - Z_I T_B)Y_I}{3C_D \rho_L Q + Q_{LK}}} \quad (13)$$

where Q and Q_{LK} are the liquid flow rate and leakage flow rate through the stage gap between the impeller and diffuser, Z_I is the impeller blade number, T_B is the blade thickness, Y_I is the representative impeller height. C_D is the drag coefficient, which can be calculated using following formula,

$$C_D = \begin{cases} C_{D,0}(1 + 0.55Sr^2) & Re > 50 \\ C_{D,0}(1 + 0.3Sr^{2.5}) & Re \leq 50 \end{cases} \quad (14)$$

where U and v are phase velocities. Sr is Strouhal number and defined by

$$Sr = \frac{d_B \Omega}{|U - v|} \quad (15)$$

$C_{D,0}$ is the drag coefficient proposed by Clift et al. (1978) without consideration of shear effect as below:

$$C_{D,0} = \frac{24}{Re}(1 + 0.15Re^{0.687}) \quad (16)$$

By equating Eqs. (11) and (12), one can calculate the critical λ_G that triggers the flow pattern transition from bubbly flow to intermittent flow, which can also be regarded as the boundary of stable and unstable flow regimes. A proper initial guess of V_{SR} and several iterations are needed to converge the solution.

7.3. Intermittent flow to segregated flow

When the transition from intermittent flow to annular flow occurs, the momentum exchange term can be neglected. Given the superficial gas velocity v_{SG} , and making a guess for the superficial liquid velocity v_{SL} , the critical liquid holdup of the film can be obtained by following procedure. From the general form of the combined momentum equation in Eq. (5), the simplified momentum balance equation can be reduced to Eq. (17) below according to segregated flow characteristics.

$$-\frac{\tau_F S_F}{H_{LF} A} - \tau_I S_I \left(\frac{1}{H_{LF} A} + \frac{1}{(1 - H_{LF}) A} \right) + (\rho_L - \rho_G) \Omega^2 R_I |_{streamline} = 0 \quad (17)$$

Based on the unified model for gas/liquid pipe flow (Zhang et al., 2003a and 2003b), the liquid holdup of the film can be calculated by

$$H_{LF} = \frac{(H_{LS}(v_T - v_S) + v_{SL})(v_{SG} + v_{SL}F_E) - v_T v_{SL}F_E}{v_T v_{SG}} \quad (18)$$

where F_E is the entrainment rate of liquid droplet to gas core, v_F , H_{LC} and v_C are expressed as

$$v_F = \frac{v_{SL}(1 - F_E)}{H_{LF}} \quad (19)$$

$$H_{LC} = \frac{v_{SL}F_E(1 - H_{LF})}{v_S - v_{SL}(1 - F_E)} \quad (20)$$

$$v_C = \frac{v_S - v_{SL}(1 - F_E)}{1 - H_{LF}} \quad (21)$$

A new value of v_F can be calculated from Eq. (17), and finally v_{SL} is obtained by

$$v_{SL} = \frac{v_F H_{LF}}{1 - F_E} \quad (22)$$

Several iterations are required to converge calculation. Thus, the transition boundary of intermittent flow to segregated flow can be determined by the curve of v_{SL} versus v_{SG} on a flow pattern map.

8. Closure relationships

In this study, the basic assumption is that the Taylor bubble prevails if the intake GVF increases further after the formation of gas pocket. According to the visualization experiments (Estevam, 2002; Thum et al., 2006; Verde et al., 2017), where the elongated bubble followed by a short liquid slug was observed, such assumption is reasonable due to the neglect of gravitational force. The governing equations have been derived in previous sections. Additional closure relationships, such as friction factors at solid wall and liquid-gas interface, liquid entrainment rate, slug liquid holdup, translational velocity and slug length, are needed to make the governing equations solvable.

8.1. Shear stress

The shear stresses in the combined momentum equation Eq. (17) are given by:

$$\tau_F = f_F \frac{\rho_L |v_F| v_F}{2} \quad (23)$$

and

$$\tau_I = f_I \frac{\rho_C |v_C - v_F| (v_C - v_F)}{2} \quad (24)$$

where the subscripts F and I are for film and interface, respectively. Based on Blasius correlation, the friction factor f_F at the solid wall can be expressed as

$$f = C Re^{-n} \quad (25)$$

where $C = 16$, $n = 1$ for laminar flow if Reynolds number is less than 2000; $C = 0.046$, $n = 0.2$ for turbulent flow in smooth pipe if the Reynolds number is higher than 3000. The wall roughness should be accounted for in a rough channel. For Reynolds number between 2000 and 3000, the friction factor is interpolated to prevent discontinuity across the transition regime.

For the interfacial friction factor, Asali (1984) correlation, which is improved by Ambrosini et al. (1991), can be used:

$$f_I = f_G (1 + 13.8 We_G^{0.2} Re_G^{-0.6} (h_F^+ - 200 \sqrt{\rho_G / \rho_L})) \quad (26)$$

where the dimensionless variables We_G and Re_G are given by

$$We_G = \frac{\rho_G v_C^2 d}{\sigma} \quad (27)$$

and

$$Re_G = \frac{\rho_G v_C d}{\mu_G} \quad (28)$$

where d is the hydraulic-equivalent diameter in ESP impeller channel $d = 4V_I/A_{wl}$. A_{wl} and V_I are the total wall area and volume of a single flow channel in the impeller. h_F^+ is the dimensionless thickness of the liquid film, which is calculated by:

$$h_F^+ = \frac{\rho_G h_F v_C^*}{\mu_G} \quad (29)$$

where

$$v_C^* = \sqrt{\frac{\tau_I}{\rho_G}} \quad (30)$$

8.2. Liquid entrainment fraction

The liquid entrainment fraction, f_E is defined as the fraction of liquid flow rate that is entrained in the gas core as droplets. The empirical correlation proposed by Oliemans et al. (1986) using the AERE Harwell data bank (Whalley and Hewitt, 1978) is incorporated in the model.

$$\frac{F_E}{1 - F_E} = 0.003 We_{SG}^{1.8} Fr_{SG}^{-0.92} Re_{SL}^{0.7} Re_{SG}^{-1.24} \left(\frac{\rho_L}{\rho_G} \right)^{0.38} \left(\frac{\mu_L}{\mu_G} \right)^{0.97} \quad (31)$$

where

$$We_{SG} = \frac{\rho_G v_{SG}^2 d}{\sigma} \quad (32)$$

$$Fr_{SG} = \frac{v_{SG}}{\sqrt{gd}} \quad (33)$$

$$Re_{SL} = \frac{\rho_L v_{SL} d}{\mu_L} \quad (34)$$

$$Re_{SG} = \frac{\rho_G v_{SG} d}{\mu_G} \quad (35)$$

8.3. Liquid holdup in slug body

The unified model proposed by Zhang et al. (2003a) is employed to calculate the slug liquid holdup, a parameter to make the governing equations solvable.

$$H_{LS} = \frac{1}{1 + \frac{T_{SM}}{3.16[(\rho_L - \rho_G)g\sigma]^{1/2}}} \quad (36)$$

where T_{SM} is expressed as

$$T_{SM} = \frac{1}{C_E} \left[\frac{f_S}{2} \rho_S v_M^2 + \frac{d}{4} \frac{\rho_L H_{LF} (v_T - v_F)(v_M - v_F)}{l_s} + \frac{d}{4} \frac{\rho_C (1 - H_{LF})(v_T - v_C)(v_M - v_C)}{l_s} \right] \quad (37)$$

And C_E is given by:

$$C_E = \frac{2.5 - |\sin(\theta)|}{2} \quad (38)$$

where θ is the inclination angle of pipe. g in Eq. (36) can be replaced by centrifugal acceleration $\Omega^2 R_I$ in radial type ESPs,

$$H_{LS} = \frac{1}{1 + \frac{T_{SM}}{3.16[(\rho_L - \rho_G)\Omega^2 R_I \sigma]^{1/2}}} \quad (39)$$

Slug flow in a rotating ESP can be viewed as co-current downward flow and the inclination angle can be assumed to be $\theta = -\pi/2$. Thus, $C_E = 0.75$.

The mixture density in slug body is defined as:

$$\rho_S = \rho_L H_{LS} + \rho_G (1 - H_{LS}) \quad (40)$$

The friction factor is calculated with the Reynolds number given by

$$Re_S = \frac{\rho_S v_S d}{\mu_L} \quad (41)$$

An initial guess using Gregory et al. (1978) correlation can be used:

$$H_{LS} = \left(1 + \left(\frac{v_S}{8.66} \right)^{1.39} \right)^{-1} \quad (42)$$

8.4. Liquid film geometry

Assuming a uniform film thickness δ_L in ESP slug flow (Sylvester, 1987), the geometrical parameters are derived as below:

$$A_C = \pi(d_I - 2\delta_L)^2/4 \quad (43)$$

$$A_F = \pi\delta_L(d_I - \delta_L) \quad (44)$$

$$S_I = \pi(d_I - 2\delta_L) \quad (45)$$

$$S_L = \pi d \quad (46)$$

where A_C and A_F are the cross-sectional area of gas core and liquid film. S_I and S_L are the perimeter of interface and liquid film, respectively. d_I is the equivalent hydraulic diameter in impeller. The hydraulic diameter of liquid film and gas core are given as:

$$d_F = 4\delta_L(d_I - \delta_L)/d_I \quad (47)$$

$$d_C = (d_I - 2\delta_L) \quad (48)$$

Therefore, the Reynolds number for gas core and film can be determined by:

$$Re_C = \frac{\rho_C v_C d_C}{\mu_C} \quad (49)$$

$$Re_F = \frac{\rho_L v_F d_F}{\mu_L} \quad (50)$$

8.5. Translation velocity

The translational velocity was proposed by Nicklin (1962):

$$v_T = c_0 v_s + v_D \quad (51)$$

where c_0 is a flow coefficient, equal to 1.2 for turbulent flow and 2 for laminar flow, based on the liquid slug Reynolds number, as given by Eq. (41). Using Bendiksen (1984) correlation for the drift velocity, v_D , the translational velocity of Taylor bubble is given as:

$$v_T = c_0 v_M + 0.54\sqrt{gd} \cos \theta + 0.35\sqrt{gd} \sin \theta \quad (52)$$

Although Eq. (50) holds within $0 \leq \theta \leq \pi/2$, it is applied here to calculate translational velocity downward a streamline. According to Zhang et al. (2003a and 2003b), Eq. (50) has been used to calculate the translational velocity of downward slug flow in the unified model. Thus, consider $\theta = -\pi/2$ in the rotating centrifugal two-phase flow, v_T can be obtained by:

$$v_T = c_0 v_M - 0.35\sqrt{\Omega^2 R_I \sin \beta_I d} \quad (53)$$

where β_I is the representative blade angle at the impeller outlet.

9. Results and discussions

This section first presents a demonstration on how the comparison of mechanistic model predictions with experimental data was conducted. By applying similar analysis procedure, further validations of model predicted flow patterns against corresponding experimental results are illustrated later.

9.1. Comparison of model predictions

As pointed out by Gamboa (2008), the changes of flow patterns inside the rotating ESP coincide with the deflection points on ESP performance curves. Thus, by searching the deflections on H - Q performance plots, different flow patterns inside ESP impeller can be summarized by such indirect observation manner.

Fig. 6 presents a typical comparison of model predicted flow patterns in the rotating ESP impeller against that detected from experimental results. The flow conditions for the selected case are at $P_{sep} = 150$ psig, $N = 3500$ rpm. As seen from the dashed-lines in Fig. 6(c), the transition boundaries calculated by mechanistic model also divide the flow pattern map into four regimes, namely dispersed bubble flow (I), bubbly flow (II), intermittent flow (III), and segregated

flow (IV). The colored markers correspond to different flow patterns, which are summarized from Fig. 6(a) and (b).

Intuitively, the model predicted transition boundaries are consistent with literature visualization experiments in Fig. 2 (Gamboa, 2008). At low Q_G , the dispersed bubble flow regime encompassed by the blue line or bubbly flow regime bounded by the red line dominates. With increase of gas flow rate or decrease of liquid flow rate, the slug flow takes place with the boundary depicted by black dot-dash line. The segregated flow (IV) occurs at relatively low Q_L or high Q_G . Compared to the flow patterns detected from ESP performance curves in Fig. 6(a) and (b), the mechanistic model is found to well capture the transition boundaries.

9.2. Validations with different flow conditions

By applying similar analysis procedure, additional validations of the proposed model under various flow conditions can be conducted. The results are presented in Figs. 7–9, corresponding to $P_{sep} = 50, 100, 150$ psig, respectively.

At $N = 3500$ rpm, $P_{sep} = 50$ psig (Fig. 7(a)), the mechanistic model predicted flow patterns are comparable to that detected from experimental performance curves. At the transition boundary from dispersed bubble to bubbly flow, corresponding to the initialization of pressure surging, a good agreement is observed. It is noted that the transition from bubbly flow to intermittent flow is well predicted by the critical gas-void fraction α_G Eq. (12).

Due to the limitation of experimental facility, the measurements were only conducted at relatively low Q_G , beyond which the nil pump boosting pressure accompanied by severe fluctuations of liquid flow rates was observed. For $N = 1800$ rpm, $P_{sep} = 100$ psig, the model predicted flow patterns match the experimental detections as well, which indicates that the mechanistic model is able to account for ESP rotational effect.

For higher P_{sep} in Figs. 8 and 9, the mechanistic model can capture the transition boundaries of flow patterns in rotating ESP. The good agreement of the model predictions under various flow conditions further verifies its applicability and accuracy. The proposed model is valuable by identifying different flow regimes, under which ESPs may or may not suffer from severe performance degradations and potential dangers of pump failures. Although the proposed model is validated by limited experimental results in this paper, it is still meaningful due to the modeling framework being established, based on which future improvement can be made by incorporating better closure relationships and more comparison with experimental data from various ESPs.

Fig. 10 shows another comparison of model predicted transition boundaries against the literature experimental data from Gamboa (2008). In his study, a mixed-type ESP (GC6100) is tested. The flow pattern map is obtained by similar method as aforementioned. In Fig. 10, the flow conditions are: $N = 2400$ rpm, $P_{in} = 150$ psig, $T = 90^\circ\text{F}$. The solid lines are from experimental data, while the dashed-lines correspond to model predictions. The horizontal/vertical axis represent the normalized gas/liquid rate, which is defined by the flow rate divided by the maximum open flow rate of an ESP. As can be seen, the transition boundaries calculated by mechanistic model agree well with that from experimental data, which further validates the applicability of the proposed model in this study.

10. Conclusions

In this paper, an analytical study on ESP two-phase flow mechanisms with the quantitative mechanistic model to predict the transition boundaries of flow patterns is proposed. Based on the comparison of model predictions against experimental data, the concluding remarks can be drawn as below.

- 1) From multiphase flow mechanisms, the proposed mechanistic model is able to predict the transition boundaries of two-phase flow

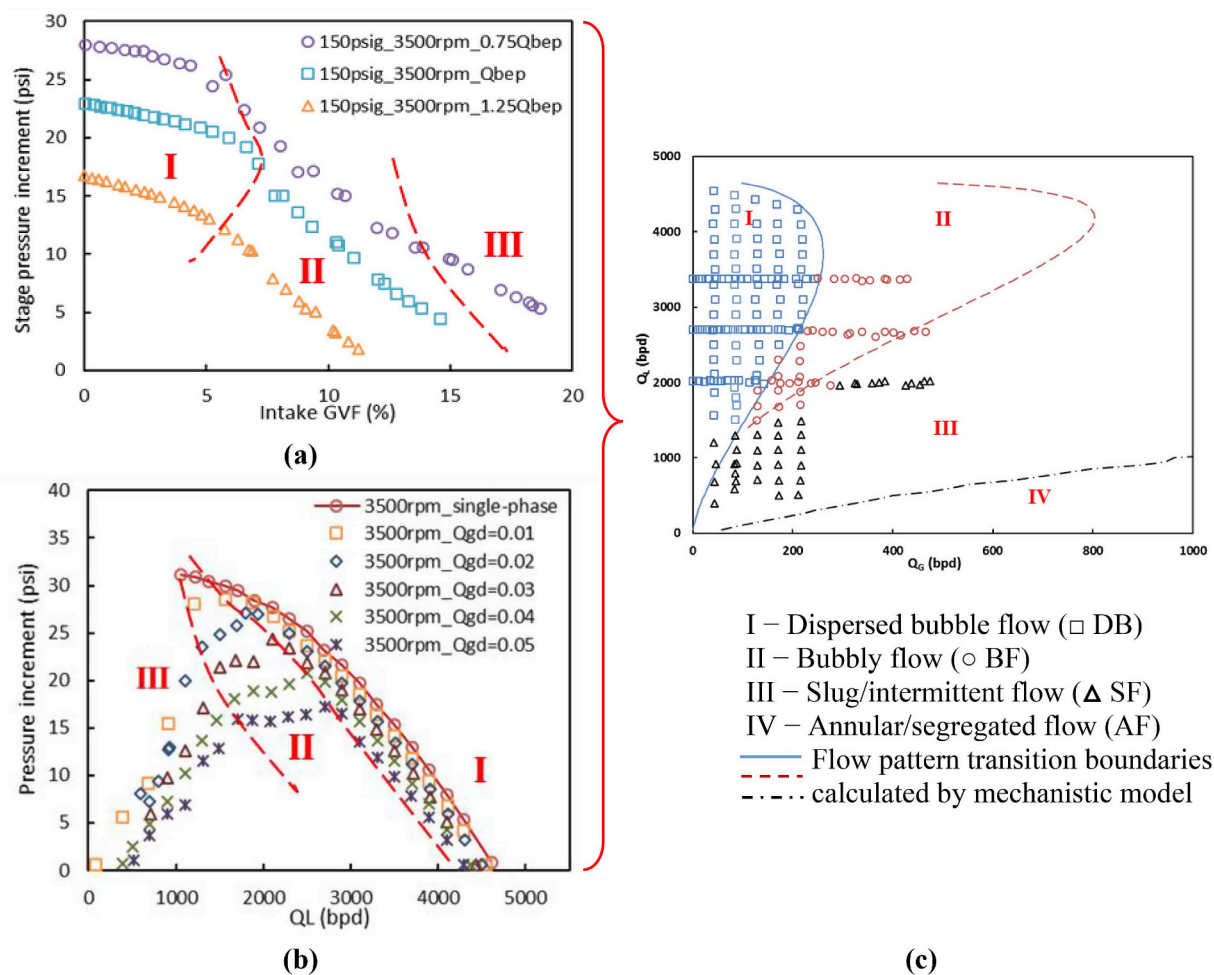


Fig. 6. A typical comparison of model predicted flow patterns with that detected from ESP performance curves at P_{sep} = 150 psig, N = 3500 rpm, (a) surging test results, (b) mapping test results, (c) flow pattern map.

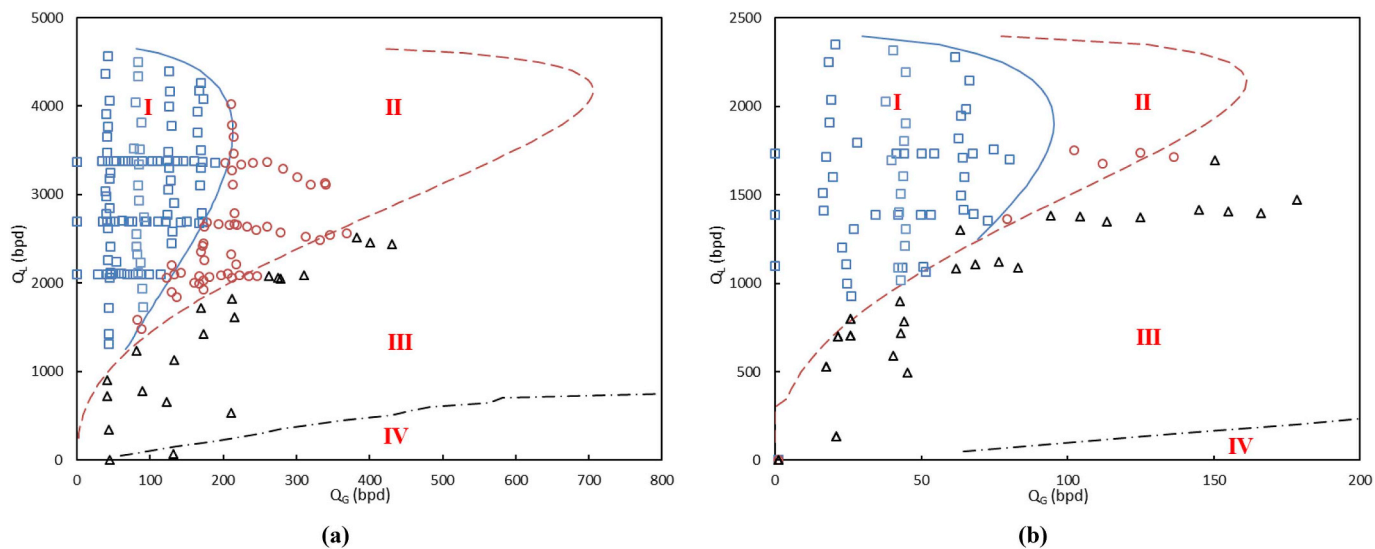


Fig. 7. Air-water two-phase flow pattern map in ESP predicted by mechanistic model at P_{sep} = 50 psig, □: dispersed bubble flow, ○: bubbly flow, △: intermittent flow, (a) N = 3500 rpm, (b) N = 1800 rpm.

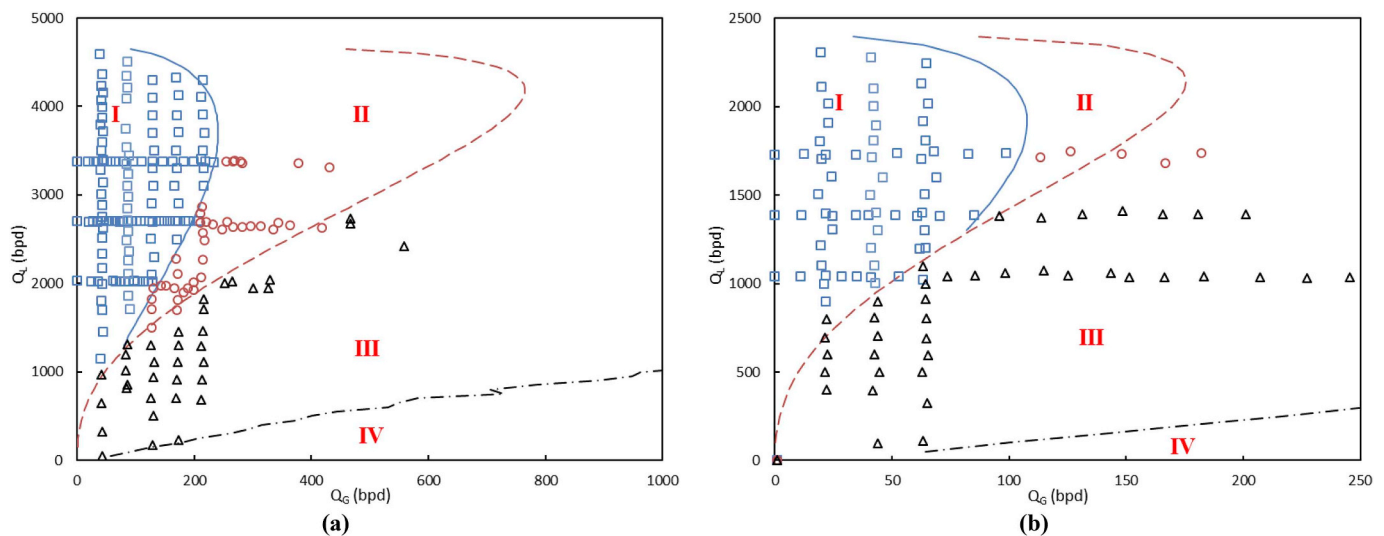


Fig. 8. Air-water two-phase flow pattern map in ESP predicted by mechanistic model at $P_{sep} = 100$ psig, $\square$: dispersed bubble flow, $\circ$: bubbly flow, $\triangle$: intermittent flow, (a) $N = 3500$ rpm, (b) $N = 1800$ rpm.

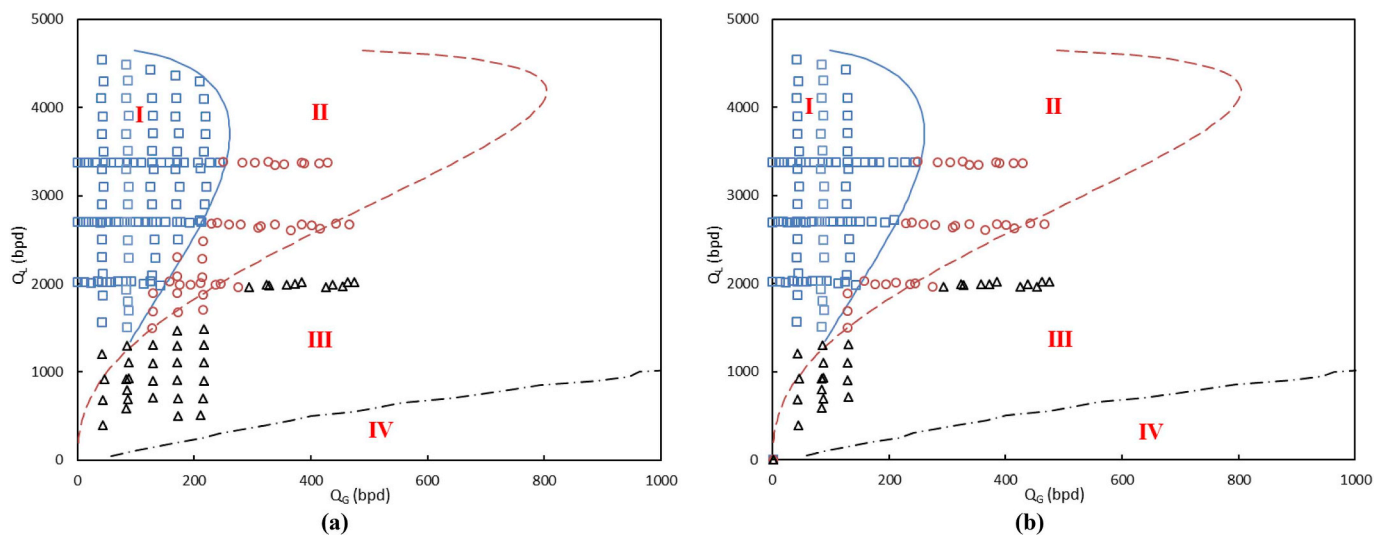


Fig. 9. Air-water two-phase flow pattern map in ESP predicted by mechanistic model at $P_{sep} = 150$ psig, $\square$: dispersed bubble flow, $\circ$: bubbly flow, $\triangle$: intermittent flow, (a) $N = 3500$ rpm, (b) $N = 1800$ rpm.

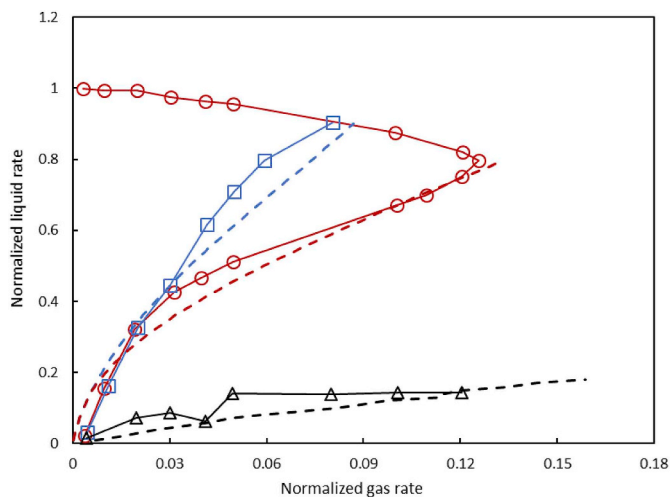


Fig. 10. Comparison of model predicted flow pattern map with Gamboa (2008) experimental data (GC6100 ESP, $N = 2400$ rpm, $P_{in} = 150$ psig, solid line: experimental data, dashed-line: model prediction).

patterns in ESPs, which plots the flow pattern map into different flow regimes, including dispersed bubble, bubbly flow, intermittent and segregated flow.

- 2) Due to visualization difficulties, an indirect method similar to Gamboa and Prado (2011a, b) is employed in this study to obtain the flow pattern map in a rotating ESP with gas-liquid flow by diagnosing the deflection points on H - Q performance curves.
- 3) Compared to the experimental data from this study and Gamboa (2008), the predicted flow pattern map demonstrates good agreement with that detected from ESP performance curves, which validates the proposed mechanistic model.
- 4) Future improvement can be made by incorporating better correlations, such as the bubble size and drag coefficient in a rotating ESP, as well as the critical gas fraction corresponding to bubbly flow and intermittent flow etc.

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Nomenclature

A	area, m ² or constant number
B	blade thickness, m
B	constant number
BEP	best efficiency point
C	absolute velocity, m/s or constant number
C_D	drag force coefficient
D	bubble diameter, m
D	diameter, m or constant number
D_i	impeller diameter, m
F	friction factor
F_E	liquid entrainment factor
$\vec{F}$	interfacial force vector, N
$\vec{g}$	gravity acceleration vector, m/s ²
GVF	gas volumetric fraction
H	channel height, m or hydraulic head, m
H	liquid holdup
L	channel length, m
$\dot{m}$	mass flow rate, kg/s
M	momentum transfer term per unit volume, Pa/m
M_a	molecular mass, kg/mol
N	phase number
N	rotational speed, rpm
P	pressure, Pa
ΔP	stage pressure increment, Pa
P	pressure, Pa
Q	flow rate, m ³ /s
Q	mass flow rate, kg/s
R	radius, m
Re	Reynolds number
S	streamline, m
Sr	Strouhal number
T	time, s
U	peripheral velocity, m/s
V	velocity, m/s
V	volume, m ³
V_o	volume, m ³
W	relative velocity in ESP, m/s
We	Weber number
X	mass fraction or mole fraction
Y	channel height, m
Z	blade number compressibility factor

Greek Symbols

A	gas void fraction or constant number
B	tangential blade angle, deg or constant number
Δ	Film thickness, m
λ_G	no-slip gas void fraction (GVF)
μ	dynamic viscosity, Pa·s
μ_t	turbulent viscosity, Pa·s
ω	specific dissipation rate or turbulent frequency, s ⁻¹
Ω	angular speed, rad/s
ρ	fluid density, kg/m ³
Σ	surface tension, N/m or slip factor
T	external stress exerting on bubbles, Pa
Γ	constant number

Subscripts

1	inlet
2	outlet
32	Sauter mean diameter
B	bubble or blade
B	blade
B_{ep}	best efficiency point
C	continuous phase

<i>C</i>	gas core
<i>CRIT</i>	critical
<i>D</i>	diffuser or dispersed phase
<i>D</i>	diffuser
<i>Eff</i>	effective
<i>E</i>	Euler
<i>EE</i>	effective
<i>F</i>	film
<i>FI</i>	fluid in ESP impeller
<i>FD</i>	fluid in ESP diffuser
<i>G</i>	gas phase
<i>G</i>	gas phase
<i>H</i>	hydraulic parameter
<i>I</i>	impeller or phase <i>i</i>
<i>I</i>	Interface or impeller
<i>L</i>	liquid phase
<i>Lg</i>	liquid to gas
<i>L</i>	liquid phase
<i>LF</i>	liquid film
<i>LK</i>	leakage
<i>Max</i>	maximum
<i>Min</i>	minimum
<i>M</i>	meridional direction or mixture
<i>O</i>	initial or equivalent
<i>P</i>	relative to project area or particle or phase (gas/liquid)
<i>R</i>	radius direction
<i>streamline</i>	projection on streamline
<i>S</i>	specific speed or shear or slug body
<i>SD</i>	diffuser channel wall
<i>SG</i>	superficial gas
<i>SI</i>	impeller channel wall
<i>SL</i>	superficial liquid
<i>SR</i>	shear in the radial direction
<i>Sep</i>	separator
<i>U</i>	peripheral direction
<i>W</i>	water
<i>W</i>	wall

Appendix-A: Calculation of Gas Density

In this study, the gas density was calculated based on Comité International des Poids et Mesures (CIPM-81) formulas (Davis, 1992). The CIPM-81 correlations require air temperature, pressure, relative humidity (or dew-point temperature) and mole fraction of carbon dioxide, as well as a number of constants. The presented ESP performance was at stage 3, whose inlet pressure and temperature were used for calculating the air density.

The density of moist air can be calculated by,

$$\rho = \frac{pM_a}{ZRT} \left[1 - x_v \left(1 - \frac{M_v}{M_a} \right) \right] \quad (\text{A.1})$$

where p is the pressure, T the temperature, x_v the mole fraction of water vapor, M_a the molar mass of dry air, M_v the molar mass of water, R the molar gas constant, and Z the compressibility factor. M_a is calculated by an auxiliary equation,

$$M_a = 28.9635 + 12.011(x_{CO_2} - 0.0004) \quad (\text{A.2})$$

where x_{CO_2} is the mole fraction of carbon dioxide. x_v is calculated by following steps. First the saturation vapor pressure p_{SV} is given by,

$$p_{SV} = 1 \times \exp \left(AT^2 + BT + C + \frac{D}{T} \right) \quad (\text{A.3})$$

where A , B , C , D are constants, as summarized in Table A.1. Next the enhancement factor f is obtained by,

$$f = \alpha + \beta p + \gamma t^2 \quad (\text{A.4})$$

where t is the temperature in degrees Celsius. The Greek letters, α , β , γ are constants as listed in Table A.1. Thus, the x_v can be calculated,

$$x_v = hf(p, t) \frac{p_{SV}(t)}{p} = f(p, t_r) \frac{p_{SV}(t_r)}{p} \quad (\text{A.5})$$

Then the compressibility Z is obtained:

$$Z = 1 - \frac{p}{T} [a_0 + a_1 t + a_2 t^2 + (b_0 + b_1 t)x_V + (c_0 + c_1 t)x_V^2] + \frac{p^2}{T^2} (d + ex_V^2) \quad (\text{A.6})$$

where, $a_0, a_1, a_2; b_0, b_1; c_0, c_1; d$ and e are constants as well, which are listed in Table A.1.

Table A.1 Summary for all constants in calculating air density

Parameters	Constants	Values
Saturation vapor pressure $p < SUB > SV < /SUB >$	A	10^{-5} K^{-2}
	B	10^{-2} K^{-1}
	C	
	D	10^3 K
Enhancement factor f	α	1.00062
	β	10^{-8} Pa^{-1}
	γ	10^{-7} K^{-2}
	γ	5.6
Compressibility factor Z	a_0	$10^{-6} \text{ K Pa}^{-1}$
	a_1	10^{-8} Pa^{-1}
	a_2	$10^{-10} \text{ K}^{-1} \text{ Pa}^{-1}$
	b_0	$10^{-6} \text{ K Pa}^{-1}$
	b_1	10^{-8} Pa^{-1}
	c_0	$10^{-4} \text{ K Pa}^{-1}$
	c_1	10^{-6} Pa^{-1}
	d	$10^{-11} \text{ K}^2 \text{ Pa}^{-2}$
	e	$10^{-8} \text{ K}^2 \text{ Pa}^{-2}$
	R	$\text{J mol}^{-1} \text{ K}^{-1}$
	M_a	$10^{-3} \text{ kg K J}^{-1}$
Gas constant R		8.31441
M_a ($x_{\text{CO}_2} = 0.0004$)/ R		3.48353

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